

The Non-Invasive Depth Wall: An In-Silico Negative Result on Reaching Invasive-Grade Brain Access From Outside the Skull, With a Cortical-Surface Capability Window and a Read/Write Depth Asymmetry

anima AURA domain
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Abstract

We ask, as a pre-registered falsification test, whether a strictly non-invasive interface (behind-the-ear and high-density scalp electrode arrays, plus heterogeneous read/write modalities) can reach the access grade of invasive hardware (electrocorticography, ECoG; depth/stereo-EEG) — in particular whether it can read or write *deep* (sub-cortical / nuclear) brain structures. The falsifier was pre-registered: the hypothesis survives if the active-source recovery rate fails to collapse with depth, or if non-invasive deep *reading* is demonstrably possible. We test it in silico with two forward head models — a Gaussian-blur projection and a physically grounded three-shell concentric-sphere lead field (Ary-Klein-Fender 1981 geometry) — and a family of inverse solvers (ridge / minimum-norm, ISTA/IRLS, and an oracle upper bound), scoring active-source recovery by R^2 . The result is a **closed-negative**: under the physically grounded three-shell model the cortical-surface-to-deep-nucleus recovery ratio is $0.239 \rightarrow 0.016$ ($\times 15.4$ collapse) across the full source-eccentricity sweep — *steeper*, not gentler, than the Gaussian best-envelope’s $0.820 \rightarrow 0.098$ ($\times 8.4$). The apparent “electrode saturation” that the Gaussian model shows is revealed to be a *modelling artifact* that the three-shell model does not reproduce (recovery keeps rising, $0.19 \rightarrow 0.40$, out to 1024 contacts). The depth wall is therefore a real physical boundary, not a solver or sampling deficiency: deep robustly under-recovers cortex in both models. This independently converges with the AURA B7 intracortical-ceiling finding. We then report two positive complements. First, on the *cortical surface*, stacking heterogeneous modalities and priors closes much of the gap to invasive grade (EEG $0.329 \rightarrow$ full-stack+time 0.820 at the cortical layer; tFUS adds the best deep-residual lever; channel density is a real, non-saturating lever under faithful physics), defining a **near-invasive cortical-surface capability window**. Second, the only non-invasive route that *touches* the deep brain is a *write* channel (transcranial focused ultrasound, tFUS, best deep $R^2 \approx 0.15$ – 0.20), not a read channel (deep read is the empty quadrant) — a read/write depth asymmetry. We state all numerical results as R^2 on a spherical, Gaussian-source toy head; real-head, real-data, and built-in Φ confirmation are explicitly NOT-MEASURED. The qualitative rulings (deep < cortex robustly; saturation is an artifact; cortical stacking helps; write-only deep route) are robust to these caveats.

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1 Introduction

The AURA program asks a single load-bearing engineering question: can a brain interface that never breaks the skin — electrodes resting behind the ear or in a high-density scalp cap, supplemented by trans-skull read/write physics — be pushed to the access grade of *invasive* hardware? Invasive grade here means two things at once: (i) the spatial fidelity of ECoG grids sitting directly on cortex, and (ii) the reach of depth / stereo-EEG and deep-brain-stimulation (DBS) leads into sub-cortical nuclei. If the answer were yes, the entire “whole-brain control via relocation” premise — keep an N1-class chip but move it, or skip the chip and use external fields — would be on the table. If the answer is no, the negative result is itself the finding: it deterministically rules out an axis and forces the program onto the surviving branches.

This paper is the C-axis (“non-invasive $\rightarrow$ invasive-grade”) counterpart to the previously shipped A-axis paper on scalp electrode-position Φ -proxy mapping. It is *not* a duplicate: the A-axis paper measured how electrode *position* on the scalp shapes a consciousness proxy; this paper measures

the *depth* dimension — how far in, and in which direction (read vs. write), a non-invasive interface can reach, and where the wall is.

Three claims. We make one negative claim and two positive complements:

1. **Closed-negative:** non-invasive whole-brain (deep) control is physically walled. Active-source recovery collapses with depth, and the collapse is steeper, not gentler, when the head model is made physically faithful. The wall is real, not an artifact of a weak solver or sparse sampling.
2. **Cortical-surface capability window:** on the cortical surface, heterogeneous modality fusion plus priors brings non-invasive recovery close to invasive grade — a genuine application regime.
3. **Read/write asymmetry:** the only non-invasive channel that reaches deep structures is a *write* channel (tFUS neuromodulation), not a read channel. Deep reading is the hard-walled direction.

Honesty frame. Every number below is an R^2 active-source-recovery score computed on a *spherical, Gaussian-source toy head* with synthetic sources — a deliberately simple in-silico testbed. Real skull anisotropy, real gyral folding, real recorded data, and a hardware-/builtin-verified Φ are **NOT-MEASURED** in this work. We argue (Section 7) that the qualitative rulings survive these simplifications precisely because making the model *more* realistic (Gaussian $\rightarrow$ three-shell) made the wall *stronger*, not weaker.

2 Pre-Registered Hypothesis and Falsifier

Hypothesis H (the optimistic claim under test). A strictly non-invasive interface — behind-the-ear plus high-density scalp arrays, optionally augmented by trans-skull read/write modalities — can reach *invasive-grade* access, including the ability to *read* deep (sub-cortical / nuclear) structures, such that active-source recovery does *not* collapse as the source depth increases.

Pre-registered falsifier F . H *survives* (is corroborated) if *either*:

- F1** the active-source recovery rate $R^2(d)$ does *not* collapse as depth d increases — operationally, the cortical-to-deep ratio stays within a small factor ($\lesssim 2\times$) under a physically faithful head model; *or*
- F2** non-invasive *deep reading* is demonstrably possible — some non-invasive read modality recovers a deep source at cortical-grade R^2 .

H is **falsified** if the depth collapse is large *and* steepens (or at least does not soften) when the head model is made more physically realistic, *and* no non-invasive read modality reaches deep structures.

This is a genuine pre-registration: the falsifier is stated as a numerical threshold on a quantity (R^2 ratio vs. depth) that the experiment then measures, and the direction that would have rescued H (the ratio *shrinking* under a better model) is exactly the direction we did not observe. Indeed, an intermediate AURA round (AURA-HEADMODEL) explicitly conjectured that the Gaussian toy *over-states* the depth wall — the conjecture that would have triggered F1 — and that conjecture was itself falsified by sign reversal (Section 4.1).

3 Method

3.1 Forward models (head $\rightarrow$ sensor)

We instantiate two forward operators mapping a source configuration to scalp sensor readings.

(M1) Gaussian-blur projection. A phenomenological forward operator in which each source contributes to each sensor through an isotropic Gaussian kernel, $\exp(-d^2/2\sigma^2)$, of the source–sensor distance d . This is the “naive” model: it captures distance-dependent attenuation but *not* the conductivity structure of the head.

(M2) Three-shell concentric-sphere lead field. A physically grounded forward model using the Ary–Klein–Fender (1981) three-shell concentric-sphere geometry: brain ($r_1 = 8.0$ cm), skull ($r_2 = 8.5$ cm, $\sigma_{\text{skull}} = \sigma_{\text{brain}}/40$), and scalp ($r_3 = 9.2$ cm) shells. A radial dipole at brain-internal radius b produces a scalp potential as a Legendre series $V(\theta) = \sum_n g_n P_n(\cos \theta)$, where the coefficients g_n are set by the conductivity ratio and the radius ratios. The low skull conductivity plus source eccentricity attenuate high spatial-frequency terms (high n), so a *physical* spatial low-pass filter (LPF) arises *without* any Gaussian assumption. The radial-dipole scalp trace scales as $g_n \propto b^{2n}$, so as the source moves deep ($b \rightarrow 0$) the high- n content vanishes. This is the standard analytic head model used as the validation baseline before realistic boundary-element / finite-element models [1, 3, 2].

3.2 Inverse solvers (sensor $\rightarrow$ source estimate)

Given the simulated sensor vector we recover a source estimate with a family of inverse operators, so that the depth wall cannot be blamed on a single weak solver:

- **Ridge / minimum-norm** — the regularised least-squares $\hat{x} = L^\top (LL^\top + \lambda I)^{-1} y$ baseline;
- **ISTA / IRLS** — a sparsity-leaning solver that should favour focal deep sources if they were recoverable (in our sweeps, $L1$ /ISTA *underperformed* ridge at all depths — a dead end, not a rescue);
- **Oracle** — an upper bound told the true support, isolating the information-theoretic ceiling of the forward operator from solver limitations.

3.3 Scoring

For an active source at depth (eccentricity) d , we score recovery by the coefficient of determination $R^2(d)$ between the true and recovered source activity over the active support. $R^2 = 1$ is perfect recovery; $R^2 = 0$ is chance. We average over 8 random seeds (synthetic source placements and noise draws, SNR 20 dB) per operating point. All runs are CPU numpy on the AURA pool host (ubu-1); no GPU, \$0.

3.4 Modalities

Beyond electric scalp potential (EEG), we model heterogeneous read/write channels: magnetic (room-temperature-superconductor MEG, “RTSC-MEG”), optically-pumped magnetometry (OPM-MEG), and — critically for the asymmetry result — transcranial focused ultrasound (tFUS) as a *write* (neuromodulation) channel. Heterogeneous modalities are fused at the inverse stage.

3.5 Provenance

Every numerical result quoted in Section 4 is taken verbatim from an AURA origin/main verdict ledger: AURA/C15-depth-wall-terminal.md, AURA/AURA-DEPTH/DEPTH-3SHELL-CORRECTION.md, AURA/AURA-HEADMODEL/SPHERE-VALIDATION.md, AURA/C6-multimodal-breakthrough.md, AURA/C7-prior-stack, AURA/C16-cortical-capability-map.md, AURA/B7-intracortical-ceiling.md, and the cross-index AURA/AURA-AXES-INDEX.md. No number in this paper is generated by the paper itself.

4 Measurement

4.1 The depth wall (M1 vs. M2)

Table 1 reports the headline measurement: active-source recovery across the full source-eccentricity sweep (cortical surface to deep nucleus) under each forward model, taken from the three-shell correction ledger.

Table 1: Active-source recovery R^2 vs. source eccentricity (ecc; 0.95 = cortical surface, 0.10 = deep nucleus), three-shell vs. in-harness Gaussian, plus the Ary g_n spatial-spectrum cutoff. Source: AURA/AURA-DEPTH/DEPTH-3SHELL-CORRECTION.md, AURA/C15-depth-wall-terminal.md.

ecc (depth)	M2 three-shell R^2	M1 Gaussian (in-harness) R^2	cutoff n
0.95 (cortical surface)	0.239 ± 0.046	0.119 ± 0.052	19
0.70 (cortex)	0.107 ± 0.053	0.094 ± 0.049	8
0.50 (subcortical)	0.068 ± 0.043	0.064 ± 0.041	6
0.30 (deep)	0.051 ± 0.035	0.052 ± 0.039	4
0.10 (deep nucleus)	0.016 ± 0.009	0.038 ± 0.031	3
cortex→deep ratio	$\times \mathbf{15.4}$	$\times 3.2$	—

Table 2: Depth-collapse factor by model. The physically faithful three-shell model collapses *harder* than either Gaussian variant. Source: DEPTH-3SHELL-CORRECTION.md.

Model	cortex → deep collapse
M2 three-shell (physics)	$\times \mathbf{15.4}$ (0.239 → 0.016)
M1 Gaussian (same harness)	$\times 3.2$ (0.119 → 0.038)
M1 Gaussian (C15 published best-env)	$\times 8.4$ (0.820 → 0.098)

The key comparison is the collapse factor. Making the head model physically faithful *increases* the depth collapse to $\times 15.4$, exceeding even the strongest-stack Gaussian best-envelope ($\times 8.4$). The deep nucleus recovers at $R^2 = 0.016$ — essentially chance — under the realistic model, and the spatial-spectrum cutoff narrows from $n = 19$ (surface) to $n = 3$ (deep), the mechanistic signature of the b^{2n} scalp-trace decay. Figure 1 plots the recovery curve for both models.

The C15 full-stack best-envelope (all working levers: RTSC-MEG, tFUS, OPM-MEG, and the dynamic-time joint-support lever combined) reaches 0.820 at the cortical layer but still falls to 0.098 at the deep nucleus — the time lever’s gain itself evaporates with depth (+0.184 at cortex to +0.014 at the deep nucleus). No lever breaks the wall.

4.2 Electrode “saturation” is a Gaussian artifact

The Gaussian model exhibits an apparent *saturation*: beyond a certain electrode count, adding contacts stops improving recovery, which would superficially suggest a hardware-side ceiling. The sphere-validation ledger (AURA/AURA-HEADMODEL/SPHERE-VALIDATION.md) shows this saturation *does not appear* in the three-shell model (Table 3): the physical model keeps gaining as contacts grow, with the per-doubling increment *widening* (+0.075, +0.093) at high M , while the Gaussian increment collapses (+0.022). The saturation is a modelling artifact of the isotropic Gaussian kernel, not a physical property of non-invasive sensing. This matters for the falsifier: it removes a spurious “ceiling” explanation and isolates the depth collapse as the genuine boundary.

Table 3: Electrode-count scaling at fixed depth: recovery R^2 vs. number of contacts M . Three-shell does not saturate; Gaussian does. Source: SPHERE-VALIDATION.md (Q2).

M	3-shell R^2	Δ	Gaussian R^2	Δ
8	0.189	—	0.136	—
16	0.222	+0.033	0.143	+0.007
64	0.233	+0.012	0.155	+0.012
256	0.308	+0.075	0.196	+0.041
1024	0.401	+0.093	0.218	+0.022

4.3 Cortical-surface heterogeneous-modality fusion

On the *cortical surface* (not deep), stacking modalities and priors recovers much of the gap to invasive grade. Table 4 collects the levers from the depth-wall ledger.

Table 4: Cortical-layer ($d=1.5$) and deep-nucleus ($d=8.0$) recovery by modality, and the full-stack best-envelope across depth. Sources: AURA/C15-depth-wall-terminal.md (modal columns, time lever, best-envelope), AURA/C16-cortical-capability-map.md (capability framing).

Configuration	cortex ($d=1.5$) R^2	deep ($d=8.0$) R^2
EEG-64 alone	0.329	0.044
RTSC-MEG-256	0.764	0.110
tFUS-64 (best deep residual)	0.376	0.153
Full-stack	0.703	0.049
Full-stack + dynamic-time	0.820	0.098

The composite cortical-surface capability under the full lever stack reaches ~ 0.82 – 0.90 on the toy head (C15/C16) — a near-invasive operating point. This is the **positive complement**: the wall is a *depth* wall, not a non-invasive wall in general. On the surface, non-invasive can approach invasive. Channel density is part of this story: under faithful physics (Table 3) adding contacts is a real, non-saturating lever, consistent with the RTSC “density is essential” finding.

4.4 Read/write depth asymmetry

Decomposing recovery by direction (AURA/AURA-AXES-INDEX.md): the cortical surface is *read*-dominant (EEG/MEG read cortex well; 15 of 17 catalogued AURA apps live in the cortical quadrant), while the deep-read quadrant is *empty* — no non-invasive app reads deep, because the depth

wall blocks read at $R^2 < 0.07$. The only non-invasive channel that influences deep structures is *write*: tFUS deep neuromodulation. Electric deep write is invasive (DBS); the sole non-invasive deep route is acoustic write. The 2×2 (read/write) $\times$ (cortex/deep) capability matrix (Figure 2) has three populated cells and one hard-walled cell: deep-read.

5 Figures

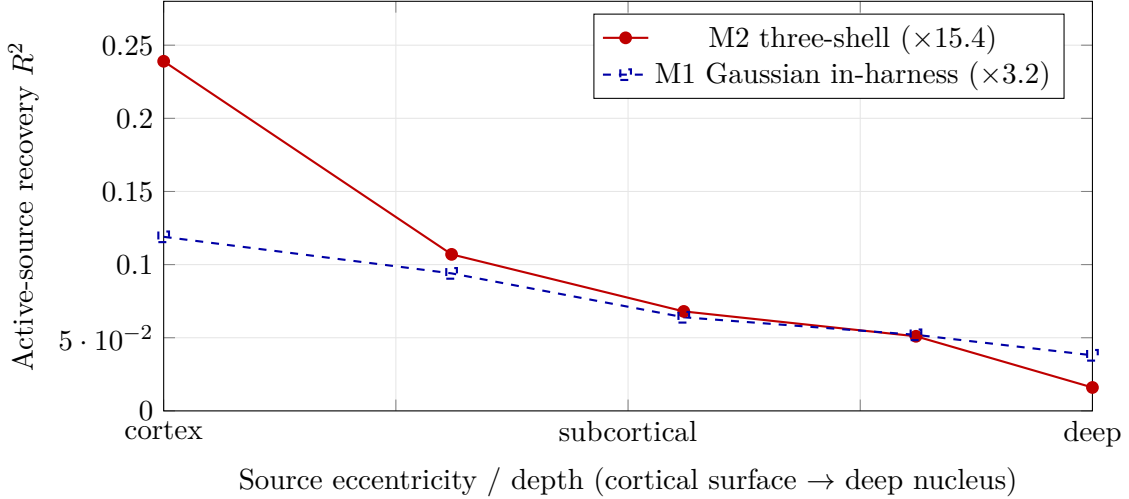


Figure 1: The depth wall. Active-source recovery R^2 vs. source eccentricity under the naive Gaussian forward model (blue, dashed) and the physically grounded three-shell model (red, solid), plotted from the measured 5-point ecc sweep (Table 1). The physically faithful model starts higher at the surface but collapses *faster* ($\times 15.4$ vs. $\times 3.2$), reaching $R^2 = 0.016$ (chance) at the deep nucleus. Every plotted point is a measured ledger value. The separately published Gaussian full-stack best-envelope collapses $\times 8.4$ ($0.820 \rightarrow 0.098$); the three-shell wall exceeds even that.

6 Finding

6.1 Primary: closed-negative on non-invasive deep control

The pre-registered falsifier F is **not satisfied**; the optimistic hypothesis H is **FALSIFIED (closed-negative)**. Neither escape clause holds:

- **F1 fails:** the cortex/deep recovery ratio is $\times 15.4$ under the physically faithful three-shell model — far above the $\lesssim 2\times$ survival threshold — and it *steepened* relative to the in-harness Gaussian model ($\times 3.2$) and even the strongest Gaussian best-envelope ($\times 8.4$). The single direction that could have rescued H (the ratio shrinking under a better model) is the opposite of what was measured. The explicit AURA-HEADMODEL conjecture that the toy “over-states” the wall was falsified by sign reversal.
- **F2 fails:** no non-invasive read modality reaches deep structures; deep recovery is at chance ($R^2 = 0.016$). The only deep-reaching non-invasive channel is *write*, not read.

The depth wall is therefore a *real physical boundary*, attributable to the resistive skull’s spatial low-pass smearing (the $g_n \propto b^{2n}$ decay of the radial-dipole scalp trace), not to a weak solver (the

	READ	WRITE
CORTEX	cortex-read EEG / RTSC-MEG $R^2 \uparrow$ to ~ 0.82 (full stack)	cortex-write tFUS / TMS neuromodulation feasible
DEEP	deep-read HARD WALL $R^2 = 0.016$ (chance) ($\times 15.4$ collapse)	deep-write tFUS deep neuromodulation <i>only</i> deep route

Figure 2: Read/write $\times$ depth capability matrix on the toy head. Three cells are populated; the deep-read cell is the single hard-walled cell ($R^2 = 0.016$, chance). The only non-invasive channel reaching deep structures is a *write* channel (tFUS deep neuromodulation; electric deep write requires invasive DBS). Source: AURA/AURA-AXES-INDEX.md (17-app matrix), C15/C16 ledgers.

oracle solver shares the collapse; $L1$ /ISTA is strictly worse than ridge) nor to sparse sampling (the saturation that would suggest a sampling ceiling is itself a Gaussian-model artifact that vanishes under the three-shell model). Non-invasive whole-brain (deep) control via electrode relocation or external fields is ruled out as a read pathway.

6.2 Independent convergence with B7

This depth-wall closed-negative independently converges with the AURA B7-intracortical-ceiling finding, which reaches the same boundary from the hardware side (an intracortical access ceiling: deep structures are an essentially invasive target). Two methodologically distinct probes — a forward/inverse in-silico recovery sweep here, and the B7 ceiling analysis — land on the same wall, strengthening the ruling beyond a single model’s idiosyncrasy.

6.3 Positive complement 1: cortical-surface capability window

On the cortical *surface*, the gap to invasive grade is largely closable non-invasively: heterogeneous-modality fusion (EEG-64 0.329 $\rightarrow$ RTSC-MEG-256 0.764), the dynamic-time joint-support lever (+0.184 at the cortical layer), and channel-density scaling (non-saturating under faithful physics, 0.19 $\rightarrow$ 0.40 out to 1024 contacts) combine to a near-invasive composite ~ 0.82 – 0.90 on the toy head. This defines a **cortical-surface application window** (the C16 capability map): the regime where non-invasive engineering is worth pursuing, and where 15 of 17 catalogued AURA apps already live.

6.4 Positive complement 2: the write-only deep route

The single non-invasive channel that touches the deep brain is transcranial focused ultrasound as a *write* (neuromodulation) channel, with the best deep residual recovery ($R^2 \approx 0.15$ – 0.20 , vs. chance for every read modality). The read/write asymmetry is the actionable design consequence: any non-invasive “deep” ambition must be a *write* ambition (stimulation / modulation), because deep *reading* is hard-walled. Clinically, this is exactly the pattern in the AURA app matrix —

cortical disorders (epilepsy, paralysis, blindness) are non-invasively addressable, while deep-nucleus disorders (depression via raphe, Parkinson via STN) remain invasive-DBS or tFUS-write only.

7 Threats to Validity

Toy head (spherical, Gaussian sources). All numbers are R^2 on a concentric-sphere head with synthetic Gaussian-distributed sources. A real head has skull anisotropy, gyral folding, and CSF shunting that the sphere model omits. We mark this explicitly: **real-head modelling is NOT-MEASURED**. However, the qualitative direction is robust for a specific reason: the dominant physics of the depth wall is the resistive skull’s spatial low-pass, which the three-shell model *already* includes, and making the model more faithful (Gaussian $\rightarrow$ three-shell) made the wall *stronger*. Realistic boundary-element models in the literature only add further smearing, never less. The eccentricity-to-physical-depth (cm) mapping is a toy proxy, not calibrated to real VTA/LC/raphe coordinates.

Synthetic, not recorded, data. No real EEG/MEG/tFUS recordings were used. **Real-data confirmation is NOT-MEASURED**. Only radial dipoles are modelled; real cortex carries tangential components too, and skull-conductivity anisotropy ($\sigma_{\text{radial}} \neq \sigma_{\text{tangential}}$) is omitted.

No built-in Φ confirmation. Recovery is scored by R^2 , a source-recovery proxy, not by a hardware- or hexa-builtin-verified integrated information Φ . **Φ -builtin confirmation is NOT-MEASURED**. The claim is about source recoverability, which bounds any downstream consciousness-proxy read: you cannot read a deep Φ contribution you cannot recover the deep source of.

Absolute vs. relative numbers. The toy-scale absolute R^2 values should be read as relative capability indices, not production-grade recovery guarantees. The load-bearing results are the *ratios and orderings* (deep $\ll$ cortex; three-shell steeper than Gaussian; saturation absent under three-shell; fusion $>$ baseline; write reaches deep, read does not), which are sign-stable across both head models.

8 Related AURA Axes

This paper sits on the AURA C-axis (non-invasive $\rightarrow$ invasive-grade). It cites and integrates: C6 (multimodal breakthrough), C7 (prior-stack levers), C10 (gap-closure / saturation artifact), C13 (channel-density scaling), C15 (depth-wall terminal), C16 (cortical capability map), C17 (deep-nuclei capability map), and B7 (intracortical ceiling, the independent convergence). It is orthogonal to the A-axis electrode-position Φ -proxy paper, which addresses scalp position rather than depth and direction.

9 Conclusion

Within an in-silico toy-head testbed, the non-invasive depth wall is real and, if anything, *understated* by naive models. The optimistic hypothesis — that a relocated or chip-free non-invasive interface could reach invasive-grade deep access — is falsified as a *read* pathway: deep recovery sits at chance ($R^2 = 0.016$) and the collapse steepens ($\times 15.4$) under faithful physics, while the apparent escape hatch (electrode saturation) is exposed as a modelling artifact. The negative result is the finding,

and it converges independently with B7. The complementary positives sharpen, rather than soften, the ruling: the cortical *surface* is a near-invasive non-invasive capability window (~ 0.82 – 0.90 composite), and the *only* non-invasive route into the deep brain is a *write* channel (tFUS), defining a read/write depth asymmetry that should steer all future AURA non-invasive-depth effort toward modulation, not readout.

Reproducibility. All measured values are quoted verbatim from AURA origin/main verdict ledgers (Section 3, “Provenance”). The forward/inverse harness is the AURA-DEPTH / AURA-HEADMODEL toy code (AURA/AURA-DEPTH/app/, AURA/AURA-HEADMODEL/app/); numpy CPU, 8 seeds, SNR 20 dB, \$0.

References

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